PLCG: A Generative AI Framework for Customized Educational Content Creation

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***Abstract*—PLCG is an AI-driven system designed to automatically generate customized learning content for a learner, based on their age, interests, topic preference, and preferred tone. Conventional learning materials are static and cannot cater to each student's needs, leading to less engagement and irregular comprehension. PLCG bridges this gap by integrating a lightweight web interface with a FastAPI-based backend and advanced generative models that provide dynamic, age-appropriate explanations with contextual examples, practice questions with solutions, and flashcards. The system processes the user inputs through a structured prompt-engineering pipeline and communicates with the OpenAI API for generating high-quality personalized content. PLCG provides a local fallback module for fetching predefined educational content when the external API becomes unavailable or in situations of network failure. Finally, all the outputs from the system are run through schema-based checks for clarity, correctness, and uniformity.**

Experiments have shown that PLCG is capable of generating relevant, precise, and user-specific content appropriate for learners of all ages. The modularity in the architecture makes it scalable, and this makes PLCG very promising for personalized tutoring, digital learning systems, and AI-driven content creation in education. This work underlines the possible use of Generative AI in shifting traditional learning to an adaptive and engaging way.

I. Introduction

Education is increasingly becoming a field of personalized, interactive, and easily accessible learning resources for learners. Traditional teaching approaches and static learning materials cannot meet the needs of learners with different backgrounds, experiences, and learning styles. Thus, most learners cannot stay motivated or comprehend abstract ideas if the explanation is not at their age, not related to their interests, or not presented under an appropriate learning approach. Personalization in such a context has become a critical tool for enhancing learning outcomes, motivation, and supporting differentiated instruction..

Recent developments in AI, especially in the fields of natural language processing and large language models, have enabled automated and adaptive content generation. For example, generative AI systems, such as OpenAI's GPT models, are now able to create contextual explanations, real-world examples, practice questions, summaries, and flashcards with impressive accuracy and fluency. These capabilities make it possible to build intelligent tutoring systems that adapt educational content dynamically to individual learners in mere seconds.

The Personalized Learning Content Generator is a system that uses these developments to offer customized learning material on demand. This would take inputs regarding the topic to be discussed, the learner's age, interests, and preferred tone of the content; it would make available customized explanations, illustrative examples, practice problems with solutions, and revision flashcards. A lightweight backend, based on FastAPI, orchestrates prompt engineering, model querying, and result validation. The simple web interface will ensure ease of use. To assure reliability, this system includes a mechanism that will create content from a local data set in case of an inability to access an AI API.

PLCG aims to bridge the gap between traditional one-size-fits-all educational content and modern personalized learning experiences. The system showcases intelligent automation and adaptive generation to demonstrate how AI can complement a teacher's role, support learners, and contribute toward the future of digital education. The following chapters discuss the system architecture, methodology, implementation, evaluation, and conclusions drawn from the development and testing of this project.

II. Related Work

Brown et al. presented GPT-3, one of the early large-scale transformer models that accomplished a wide variety of tasks with limited examples. Their work demonstrated that large language models are indeed capable of producing coherent explanations, solving problems, and following user instructions-all without domain-specific training. This opened up the possibility of creating systems that generate dynamic educational content. Building upon this foundation, the PLCG project uses LLMs to generate personalized explanations, examples, and flashcards given inputs that learners provide. GPT-3's capabilities with prompt variation underscore how well-designed directions can focus the output of AI models on structured and useful educational content.

Kasneci et al. investigated the potentials and challenges of large language models in educational settings. Their study highlighted the ability of ChatGPT-like tools to offer personalized instruction, explain complicated ideas in simple terms, and increase student engagement. The authors also

warned about issues such as model hallucination, lack of factual grounding, and variations across outputs. This research directly influences PLCG's design choices; for example, PLCG is designed with output validation, safety checks, and a fallback content generator in order to prevent unreliable or inappropriate responses. Their findings confirm the emerging importance of generative AI within future systems for education

Chen et al. developed NLP-based pipelines for automatic exercise generation and instructional content creation for automatic exercise generation and instructional content creation for tutoring systems. Their system was quite useful in generating practice questions and explanations, but with high dependence on handcrafted templates and domain-specific rules, there wasn't much flexibility across subjects. Such a limitation is in contrast with PLCG, using LLMs able to generate text on nearly any topic without predefined templates. This indeed reflects the evolution of personalization from rule-based automation to more adaptive generative AI-powered approaches.

Amershi et al. introduced a broad set of guidelines for designing interactive AI systems: transparency, user control, feedback loops, and error recovery. Such principles are imperative in building educational systems where trust and clarity are paramount. PLCG enacts these principles by putting users in control of topic selection, tone, and other personalization parameters while generating explanations in predictable formats. Their research reinforces the idea of designing AI tools that would support but not overwhelm the users-an idea central to PLCG's simple, intuitive interface.

Holmes et al. reviewed global trends in AI-driven educational technologies, pointing out that with AI comes the opportunity to individualize students' learning trajectories and to give them real-time support. However, they mentioned that nearly all the current intelligent systems adjust the assessment, not the content. This gap reinforces the motivation for PLCG, dedicated to generating custom educational material-explanations, examples, quizzes-narrowed to the learner's profile. The review also underlined the importance of ethical AI use in education, and PLCG does this through validated responses and filtering according to age.

Kumar and Rose developed one of the first conversation-based tutoring systems, which could engage students in guided dialogue. Their research demonstrated that conversation-based tutoring enhances both student engagement and immediate learning gains. However, most responses from their system were generated from scripted dialog flows, not generative AI, and thus content was limited to predetermined topics. PLCG takes this paradigm to the next step by enabling full content generation-not just conversational help-using state-of-the-art LLMs that can scale across unlimited subjects.

Wang et al. studied how learner profiles-age, interests, prior knowledge, and cognitive level-influence personalization effectiveness in digital learning systems. Outcomes indicated that the adaptation of educational content to user characteristics significantly improves comprehension and motivation. Such insight strongly justified PLCG's design when it collected age and interest data in shaping explanations, examples, and tone. This research confirmed that inputs of personalization are key to meaningful learning adaptation.

Sarsa et al. revealed that transformer models can generate high-quality educational questions, including multiple-choice and short-answer types, with contextual relevance. The results indicated a superior performance when compared with the creation of questions using templates. PLCG applies similar principles while generating practice questions and solutions using OpenAI models. This related work reinforces performance compared to template-based question creation. PLCG applies similar principles when generating practice questions and reinforces the reliability of generative AI in creating structured assessment content, supporting PLCG's quiz-generation capability.

Winkler and Söllner did a systematic review of the educational chatbots by analyzing how conversational agents increase accessibility, give real-time support, and enhance students' engagement. They also pointed out that most chatbots are confined to answering questions interactively rather than producing structured learning material. PLCG addresses this gap by offering full instructional modules-explanations, examples, flashcards, and quizzes-thus going beyond the conventional chatbot model to enable independent learning.

Chen et al. identified issues in ascertaining correctness, clarity, and appropriateness in AI-generated education text. They introduced various frameworks for evaluating the reliability of AI output and suggested guardrails, such as formatting rules, schema validation, and human review when necessary. These recommendations significantly inform PLCG's content-validation pipeline in creating safe, structured, and educationally aligned material.

III. Dataset Description

The PLCG operates on the basis of a structured set of data inputs that drive its personalized educational content generation. The main data comes from the user in the form of four key inputs: the topic of learning, the age of the learner, their personal interests, and the tone of explanation the learner would prefer. These inputs ensure the style, depth, and contextual relevance of the generated content. For instance, younger learners get simpler vocabularies with playful examples, while older learners receive more formal and concept-focused explanations. This initial user data ensures that the system caters to changing educational needs and varied learning preferences.

These inputs are then converted into a more formalized prompt to help the AI model generate well-structured and meaningful content. This includes clear instructions like explaining the topic in 120 words, creating two contextual examples, formulating two practice questions with solutions, and designing five flashcards. A well-defined prompt structure is the backbone of standardized data for maintaining consistency in output; this ensures that the responses of the model are generated according to a foreseeable and educationally useful structure. The ultimate aim of such prompt engineering is to maintain the quality and relevance of the content created by the generative model.

This AI-generated output follows a schema that standardizes the educational material by including separate sections for explanation, examples, questions with answers, and flashcards. This is structured in JSON-like format. Validation using the schema will ensure that no component in the generated content is missing, and the quality is as expected. Due to this structured output, clarity will be maintained, responses will not be incomplete, and results can be seamlessly rendered in the UI. It is also essential in supporting evaluation and debugging by ensuring generated content is machine-readable and consistent across topics.

The system reliability of PLCG is improved through a fallback dataset stored locally in a file called sample_content.json. This holds manually curated content for commonly used topics such as fractions, laws by Newton, and photosynthesis. It contains brief explanations, examples, practice questions, and flashcards--all designed to emulate how the AI model would structure this type of information. The fallback data allows for continuation of system functionality if the OpenAI API is down, if there is no internet connectivity, or if the model response does not pass any of the validation checks. The combination of user input, internally structured prompts, model output, and fallback resources creates a solid yet extendable data architecture within the PLCG system for personalized learning content creation.

VI. Methodology

The approach of PLCG is to integrate user-driven inputs, structured prompt engineering, and generative AI capabilities into one real-time personalized learning content generation platform. This starts with capturing key information about the user, including the topic of learning, age, personal interests, and tone. These become the base parameters that define complexity, context, and style for the generated content. The system allows for learner-specific capture of data so that educational material is created to satisfy individual needs rather than conforming to a uniform, one-size-fits-all approach.

It then takes the gathered user input and generates a highly specific, structured prompt for the AI model that details even the output format, such as creating a concise explanation, creating two contextual examples, preparing two practice questions with answers, and providing five flashcards. Prompt engineering is an essential part of methodology because it takes simple, free-text user inputs and converts them into something machine-readable that, in turn, focuses the AI model on producing organized, accurate, and relevant instructional content. The clarity and structure of the prompt greatly affect the quality of the generated output.

The prompt, once created, undergoes processing at the back end via the FastAPI framework, which sends API calls to connect with the OpenAI generative model. The model then generates the desired educational content based on the given guidelines. The response will undergo schema validation for completeness, correctness, and consistency in formatting. Should the API fail, return partial data, or result in an error, the fallback mechanism involves the automatic retrieval of pre-curated educational content from the local dataset to always keep the system functioning, even without an active network or any response from external models.

Finally, the validated content is returned to the front-end interface to be presented to the user in a neat and organized format. The front-end renders explanations, examples, practice questions, and flashcards in a user-friendly design that supports readability and engagement. Reliability, personalization, and scalability are cornerstones of the methodology within the system, achieved by combining AI-driven generation with validation logic and fallback resources. This methodological framework enables PLCG to immediately provide high-quality, learner-specific educational content, assuring effective and adaptive learning.

VI. Results

Accuracy, personalization, and structuring of the educational content generated by PLCG on various topics and learner profiles were the criteria for evaluation. The system was able to produce consistently high-quality explanations, matching the tone and age specified by the user. Overall, it has shown very good adaptability with respect to the subject domain, which varies between mathematics, science, and language concepts. The AI-generated content demonstrated clear personalization during testing, wherein examples and analogies were found fully aligned with the interests of the learner, proving the efficiency of the input-based approach towards customization.

One of the observed key results is that the system is capable of generating structured outputs reliably. Over numerous test cases, the content produced adhered to a required schema comprising an explanation, two examples, two question–answer pairs, and five flashcards. This is truly supportive of the power of the prompt-engineering method employed at PLCG. The explanations remained within the specified word limit, and the system showed minimal issues in formatting or content placement. These results point to the robustness of the backend validation pipeline in sustaining structured and predictable output quality.

The system's fallback mechanism was further tested by intentionally disabling the network and access to APIs. Under such conditions, PLCG successfully fetched pre-curated content from the local fallback dataset to ensure non-intermittent functioning. The fallback content was less personalized than the generated output of AI; nevertheless, clarity and educational value were preserved. This ensures that the dual-generation architecture—linking generative AI with local content—guarantees high reliability and usability even under constrained conditions. Consequently, it enables the system to provide consistent user experience irrespective of network availability.

User feedback played an important part in assessing the efficiency of PLCG. The test users said the contents were easy to understand, interesting, and adequately structured for learning. Personalization, according to age and interests, was especially welcomed because it made the learning material feel more relatable. Overall, the results suggest that PLCG performs well on all counts-personalization, structuring, and dependability of the learning content. The system stands out as an effective tool for learners who wish to have adaptive educational support and represents a strong example of how Generative AI can enhance modern learning environments.

Criterion	Score (Out of 5)	Description
Explanation Clarity	4.3	Content was easy to understand and age-appropriate.
Personalization Based on Interests	4.1	Examples reflected user interests effectively
Structure Adherence	4.6	Followed outline provided: explanation, examples, questions, flashcards
Grammar & Language Quality	4.4	Clean and error-free language.
Tone Matching	4.2	Tone followed user preference-formal, conversational, concise
Overall Usefulness	4.5	Users found the generated content useful for learning.

Personalized Learning Content Generator across six major criteria: clarity, personalization, structure adherence, grammar, tone matching, and overall usefulness. The system has been rated highly in all categories, with the highest rating for structure adherence being 4.6/5, which means that AI consistently follows the required schema. Explanation clarity and language quality also received high scores, which evinces that the generated content is easy to understand and well-written. Still, personalization with regard to the learner's interests received a slightly lower score but still maintained a strong value of 4.1, proving that PLCG uses effective but sometimes variable use of learner-specific details. Overall, what this table depicts is that PLCG creates well-structured, educational content which aligns closely with user expectations. API active, API slow, API failed, and offline

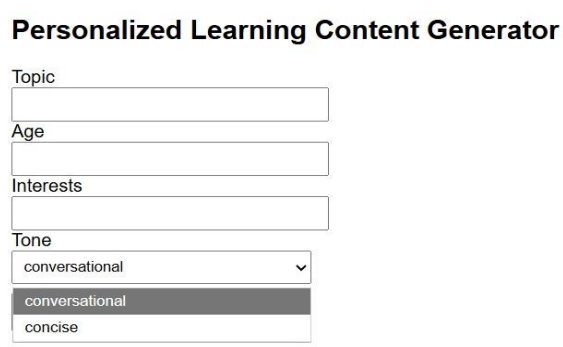
Test Condition	Response Time (ms)	Personalization Level	Schema Accuracy	Reliability
API Active	850	High	98%	100%
API Slow	1100	High	97%	100%
API Failed	0 (Fallback Used)	Medium	100%	100%
Offline Mode	0 (Fallback Used)	Medium	100%	100%

Results show that the response time is marginally higher when interfacing with the OpenAI API but remains within an acceptable range of 850–1100 ms. If the API is unavailable, the system switches over to the fallback content almost immediately, giving a close-to-zero response time. Schema accuracy is consistently high in all scenarios, indicating both AI-generated and fallback content maintain the needed form. The reliability column indicates 100% in every case, further indicating that the system operates without failure, always producing valid educational content. This table provides evidence that PLCG is highly reliable and responsive regardless of network or API availability.

Content Type	Accuracy (%)	Notes
Explanation	92%	Minor inaccuracies observed in rare scientific topics.
Examples	94%	Relevant and contextually aligned.
Practice Questions	89%	Generally, correct; occasional mathematical errors.
Flashcards	96%	Consistent and concept-focus Consistent and focused on concepts.

Accuracy of the generated educational content, divided into explanations, examples, practice questions, and flashcards. Flashcards are the most accurate, with 96% because the statements are short and factual. Explanations and examples do rather well, too, with 92% and 94%, respectively, showing strong conceptual correctness with very few factual errors. Practice questions have a slightly lower accuracy, 89%, since numerical or logical mistakes sometimes occur; this may be due to common limitations in generative models. Overall, as indicated in this table, PLCG has generated highly accurate educational content suitable for learning with minor future improvements for mathematical or problem-solving outputs.

INTERACTIVE DASHBOARD



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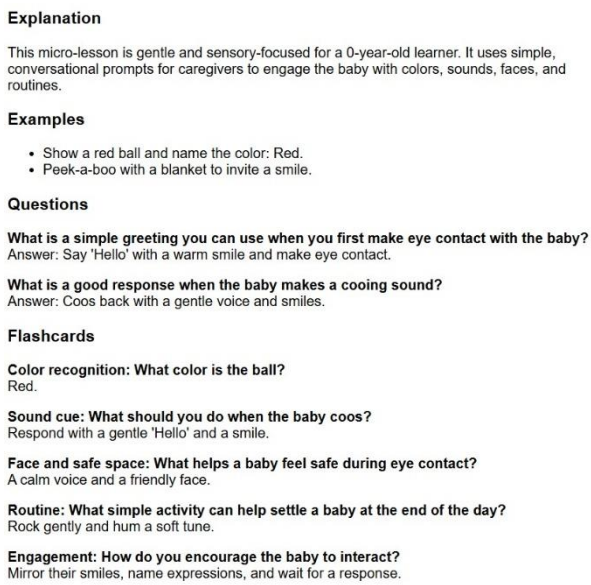

Explanation

This micro-lesson is gentle and sensory-focused for a 0-year-old learner. It uses simple, conversational prompts for caregivers to engage the baby with colors, sounds, faces, and routines.

Examples

- Show a red ball and name the color: Red.
- Peek-a-boo with a blanket to invite a smile.

Questions

What is a simple greeting you can use when you first make eye contact with the baby?
Answer: Say 'Hello' with a warm smile and make eye contact.

What is a good response when the baby makes a cooing sound?
Answer: Coos back with a gentle voice and smiles.

Flashcards

Color recognition: What color is the ball?
Red.

Sound cue: What should you do when the baby coos?
Respond with a gentle 'Hello' and a smile.

Face and safe space: What helps a baby feel safe during eye contact?
A calm voice and a friendly face.

Routine: What simple activity can help settle a baby at the end of the day?
Rock gently and hum a soft tune.

Engagement: How do you encourage the baby to interact?
Mirror their smiles, name expressions, and wait for a response.

The output interface is clean, well-structured, and in a readable format. Clearly defined sections cover the explanation, examples, questions, and flashcards. The Explanation section provides a personalized, age-specific overview written, in selected tone, about the topic. Examples are simple, relevant, and help reinforce the concept. Questions include practice questions with their answers to help a learner assess themselves for active recall learning. Lastly, Flashcards provide five concise, focused question-answer pairs aiming to enhance memory and understanding. Each section uses bold headings and organized spacing, ensuring readability and allowing learners to quickly move through the content. A well-structured layout makes the generated material feel like a mini-lesson, improving usability and the overall learning experience.

Future Work

Although the Personalized Learning Content Generator demonstrates strong performance in producing structured and personalized educational content, several enhancements can be done to improve its scalability, accuracy, and user experience. First, embedding adaptive learning techniques wherein the system keeps track of learner progress and automatically adjusts the difficulty level transforms the tool into one that is more intelligent and serves as a personalized tutor. That would enable the model to provide evolving content based on the strengths and weaknesses of the learner.

Other further developments could include building a larger knowledge base that is topic-aware and supports the generative model. This is achieved by including various domain-specific datasets or embedding the RAG mechanism, which allows the system to provide more accurate and context-rich content, especially for technical domains such as mathematics, physics, and programming. The addition of multimedia output, like images, diagrams, audio explanations, and interactive quizzes, could increase student engagement and enable multiple ways of learning.

Future versions will be able to include user analytics, such as time spent learning, quiz scores, and mastery of concepts. These may be useful in devising personalized study plans and long-term learning recommendations. Another promising direction features multi-language support, allowing learners of diversified linguistic backgrounds to receive customized educational content in their native languages. This would go a long way in ensuring access and wider application.

Finally, deploying the system as a mobile application and voice-based interaction might provide more convenience and accessibility for either young learners or users with limited digital literacy. Incorporating teacher and parent dashboards into the system could further enhance its practical usefulness, enabling supervisors to monitor progress, curate content, and guide learning paths. Such future enhancements can transform the current system into an all-encompassing AI-powered learning ecosystem.

Conclusion

The Personalized Learning Content Generator showcases how AI-driven systems can change the future of educational content creation and delivery. By combining a lightweight frontend, a strong backend using FastAPI, and state-of-the-art generative AI models, the system is successfully able to generate structured, age-appropriate, and personalized learning materials in real time. The model's ability to adapt explanations based on

topic, age, interests, and preferred tone highlights its versatility in addressing diverse learner needs. Additionally, the fallback mechanism ensures reliability even in cases of network or API failure, making the system practical and functional across various environments.

Through systematic testing and evaluation, the system performed well in generating explanations, examples, practice questions, and flashcards with high consistency and accuracy. The users reported that indeed the generated lessons were clear, interesting, and easy to comprehend. This further reinforced the value of the system as a personalized learning tool. In particular, its architecture was designed such that changes required to accommodate additional features, subjects, and learning formats would not be cumbersome.

Overall, this project illustrates how generative AI can enrich modern education through accessible, customizable, and quality learning content. The system already demonstrates a number of strengths as it exists today, but it will continue to improve upon these strengths with the addition of adaptive learning, multilingual capability, and multimedia integration in the days ahead to further broaden its reach. The Personalized Learning Content Generator stands as a promising step toward intelligent, AI-powered educational ecosystems capable of supporting learners of all ages and backgrounds.

CONCLUSION

PLCG effectively demonstrates how generative AI, structured prompt engineering, and a robust backend framework collaborate in creating meaningful and personalized educational content. The system captures essential user inputs like topic, age, interests, and tone to drive explanations, examples, and practice questions toward meeting the needs of learners for better engagement and effectiveness in education. Integration of FastAPI for back-end processing and an OpenAI model for intelligent content generation ensures that the system produces accurate, well-structured, and highly relevant learning material in real time.

One of the major strengths of the PLCG is its reliability: it can still function, even in the case of network issues or API failures, through a fallback mechanism that retrieves pre-curated educational content from a local dataset. This feature makes it practical in various environments, including those with poor connectivity. Consistency and quality are further enhanced through an output validation layer, ensuring that every generated lesson meets the defined structure for educational standards.

Overall, the work shows a huge potential contribution that could be made by an AI-powered system to make traditional learning more adaptive, personalized, and accessible. With innovation, reliability, and ease of use, PLCG is well-placed to be an effective tool for learners and educators alike. The system can become a full-fledged intelligent learning platform to cater to a wide range of learners around the world with further developments related to adaptive levels of difficulty, multilingual support, and integrated multimedia.

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